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Glaucoma detection using novel perceptron based convolutional multi-layer neural network classification

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Abstract
Glaucoma in retina is considered as a significant cause of irreparable vision loss. For the automated glaucoma detection on fundus images, several approaches have been recently developed. However, the extraction of optic cup (OC) boundary considered as critical work due to the blood vessels interweavement. To accurately detect glaucoma and correspondingly the optic cup OC and optic disk (OD) boundary, the Perceptron based Convolutional Multi-Layer Neural network classification with Weighted Least Square Fit holistic feature extraction has been proposed. In this study, the Glaucoma classification has been performed using Perceptron based Convolutional Multi-Layer Neural network. The optic disc and optic cup boundary is detected by entropy based estimation. After the optic disc and cup boundary are segmented, the disc ratio and holistic local features are extracted by Weighted Least square fit. The Perceptron based Convolutional Multi-Layer Neural network classification is performed for various datasets such as Drishti-GS and Rim-ONE dataset to accurately classify the Glaucoma in retinal images. The proposed method is evaluated for the OC and OD segmentation, accurate glaucoma detection in terms of accuracy, sensitivity, specificity, dice and overlap, and Jaccard parameters.

Keywords Retina blood vessels · Convolutional neural network · Classification · Glaucoma · Segmentation · Optic disc

1 Introduction

Computer aided diagnostics enhances the diagnostic speed and assisted in locating the specific area. Diabetic retinopathy (DR) occurs at the back of the eye (retina) due to the blood vessel damage caused by prolonged Diabetes mellitus. New, abnormal blood vessels

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are manufactured by retina. Neo-vascular glaucoma then occurs if the new blood vessels grow on the iris, fluid flows are then closing off in the eye and eye pressure is raised. The major challenge while detecting Glaucoma is it doesn't show any symptoms/pains and also the vision is normal at an early stage. During the advanced stage, it is noticeable when the patient has lost 70 percent of his/her vision. Hence for early glaucoma detection regular screening of eye is necessary. Basically, for DR detection the Fundus photography is widely used by ophthalmologists (Kumar et al., 2019). Ophthalmic associated ailments like glaucoma, age concerned degeneration of fovea, hypertension, diabetic retinopathy, choroidal neovascularization, and arteriosclerotic are primarily causing eye blindness i.e. in-ability to view the surrounding because of affected eye by the human beings. So, prior identification of such affected eye areas could make us adopt some preventive treatments to avoid the sudden blindness that the people of these days are exposed to Mannis (2016). Traditional general stereo categorical images (Abramoff et al., 2007) were deployed to automate the optical discs segmentation operation with the aid of some unique features to augment the efficiency of the segmentation operation. Identification of Glaucoma ailments was undertaken with the usage of super-pixel optics dependent classifying methodology (Cheng et al., 2013). This way of segmenting the optical cups and discs was able to screen the glaucoma disorders. The glaucoma example from retinal images is shown in Fig. 1.

Generally, for medical diagnosis pre-processing, feature selection, and disease classification are followed. The challenge seen in glaucoma detection is the accurate localization of affected optical disc and based on size of the cup which is either too small or large. By using retinal images, efficient techniques such as Particle swarm optimization improved ensemble deep neural networks have used in the existing studies for the optic disc OD segmentation. The single network bias has been overcome by the ensemble segmentation technique(Zhang & Lim, 2020). The two critical issues seen in optical nerve head are variant spatial layout of vessels and small spatially sparse optic cup boundaries solved by the spatial-aware joint segmentation method(Liu et al., 2019). Due to the large variation, highly weak contrast among optic head areas and heavy overlap the multi-task collaborative learning MCL-Net has been developed for the quantification of optic nerve head multi-indices. However, in some cases, it is complex to identify the OC/OD boundary with indistinct and glaucoma fundus(Zhao & Li, 2020). A Multi-label deep learning framework namely M-Net used for OC and OD segmentation issues by generating the one-stage

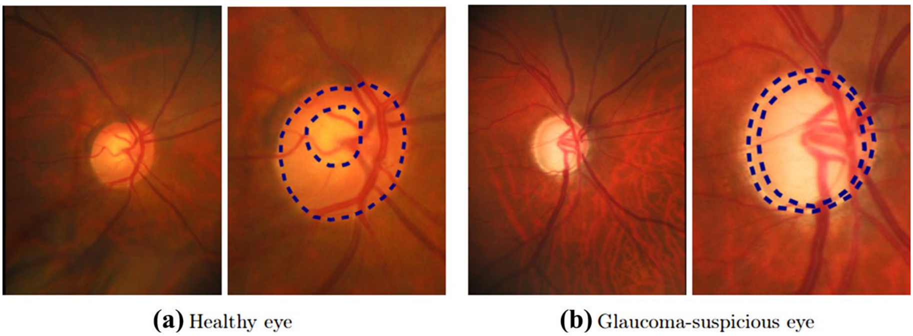


Fig. 1 **a** A healthy and **b** glaucoma eye example. **a** and **b** Right-hand picture shows the enlarged optic disc area, where outer dashed line specified the optic disc border whereas inside dashed line shows optic cup border

multi-label model. Higher accuracy resulted for vertical disc and cup compared to horizontal (Fu et al., 2018). Generally, for medical image analysis, the deep learning approaches are not used for glaucoma evaluation due to its small data size and hence this issue overcame by REFUGE-Retinal Fundus Glaucoma Challenge consists of large data with 1200 fundus images. Further, the two basic tasks such as OD/OC segmentation and classification of glaucoma have been performed (Orlando et al., 2020). The DENet-disc aware ensemble network based deep learning model has been further developed for local optic disc boundary incorporation and global fundus image integration related to the deep hierarchical context. In addition to that, the DENet gives glaucoma detection precisely even without segmentation (Fu et al., 2019).

It is possible that the optic disc and cup are not directly segmented and hence the optic disc and cup of minimum bounding boxes have been detected and evaluates as adorned ellipses inside the boxes. Joint RCNN namely Faster RCNN has been used for this process of optic disc and cup boundary boxes detection (Jiang et al., 2019). From the existing studies, the challenges seen are the extraction of optic cup boundary considered as critical work due to the blood vessels interweavement. Generally, for medical diagnosis feature selection and disease classification are followed and in this study the significant challenges seen as glaucoma detection which is the accurate localization of affected optical disc based on size of the cup which is either too small or large.

The proposed Perceptron based Convolutional Multi-Layer Neural network classification with efficient feature extraction method using Weighted least square with holistic features are executed on Drishti-GS and Rim-One datasets to overcome these challenges and the major contribution of this study includes,

- To perform optic disc OD and optic boundary OC segmentation accurately by implementing the improved CNN with entropy estimation and Cup-Disk ration CDR values measured by Weighted least square with holistic features.
- To detect the Glaucoma in retinal images by using the Perceptron based Convolutional Multi-Layer Neural network classification.
- To compare the proposed method with various existing methodologies and datasets to show the effectiveness of the current study.

The following Sect. 2 describes the proposed Perceptron based Convolutional Multi-Layer Neural network classification with efficient feature extraction method using Weighted least square with holistic features and followed by the proposed method evaluation and comparison with existing methods illustrated in Sect. 3. Finally, the paper concluded in Sect. 4.

2 Related works

For the multi-scale fundus image representation, a new framework based on Weakly Supervised Multitask Learning WSMTL based CNN was implemented and the model trained based on binary diagnoses such as pixel-level evidence map, normal or glaucoma, diagnosis prediction, and segmentation mask (Zhao et al., 2019). The second leading cause of blindness globally is Glaucoma which is not reversed and is loss of vision caused by Glaucoma. For high quality Glaucoma screening, Yanbao—user-friendly app developed based on the clinical parameters (Guo et al., 2018). Within the retinal images, the complex blood vessel structure and severe intensity inhomogeneity have been considered as severe issues and it overcame by Locally statistical active contour model (LSACM) incorporates probability information based on multi-dimensional feature (Gao et al., 2019). In fundus images, the segmentation and localization of optic disc are considered as significant steps for early retinal disease onset detection like glaucoma, diabetic retinopathy, and macular degeneration. For this process, a novel CNN is used for OD segmentation (Bhatkalkar et al., 2020). In retinal fundus images, the basic two-stage Mask R-CNN has been developed for optic nerve head localization and optic disc/cup segmentation. This study enhanced the detection with the help of new fine-tuning approach by cropping output combined with original training image using various scales. Vertical cup to disc proportion measured for glaucoma diagnosis (Almubarak et al., 2020).

Through a systematic classification pipeline, the regional classification problem is modeled from the OD segmentation problem in retinal fundus images. By textural and statistical properties the image regions are characterized in a classification framework and hence this kind of technique is potential against the OD localization corresponds to multifaceted challenges (Rehman et al., 2019). The DenseNet integrated with fully convolutional network namely FC-DenseNet designed for the semantic segmentation. Thus, OD and OC automatic segmentation performed in this study and effectiveness resulted. Also, the automated classification results are enhanced by vertical cup to disc ratio with horizontal cup to disc ratio and however, the diagnosis results can be improvised by utilizing the OD's complete profile (Al-Bander et al., 2018). OD segmentation challenges occur due to the vessel's presence and it has been expected to overcome by pre-processing techniques. These techniques used morphological operations such as histogram equalization or closing and opening. However compared with OD, OC segmentation challenges show complexity because of the optic cup interweavement with surrounding tissues and blood vessels. Thus enhanced segmentation techniques are necessary for accurate glaucoma diagnosis (Thakur & Juneja, 2018). Smoothen region of the input retinal image is identified by the Greyscale low intensity images corresponding to feature extraction. This study used the Gaussian kernel fuzzy c-means clustering for detecting the retinal image affected area in which the cup boundary is extracted. Further, the Support Vector Machine classification has been applied to identify the prior affected area (Manojprabhakaran, 2017).

In Aamir et al. (2020), a new multi-level deep convolutional neural network (ML-DCNN) model is presented for diagnosing glaucoma using retinal fundus images. In Bharati et al. (2020), an efficient hybrid deep learning model using the integration of VGG, data augmentation and spatial transformer network (STN) with CNN for lung disease diagnosis. In García et al. (2020), a novel CNN-LSTM-based model is developed to detect glaucoma from raw spectral-domain optical coherence tomography (SD-OCT) volume. In Zhao et al. (2020), an efficient curriculum learning paradigm (EGDCL) is developed for the detection of glaucoma. Some other methods are also available in the literature (Coccia, 2020; Coccia & Watts, 2020; Fatima Bokhari et al., 2018; Lazouni et al., 2019; Li et al., 2021; Liao et al., 2020; Mansour, 2017, 2018; Mansour et al., 2021; Tabassum et al., 2020; Xu et al., 2020).

In Yu et al. (2019), an effective optic disc and cup segmentation technique is presented using modified U-Net model combining the pretrained ResNet34 model as encoding layers with U-Net decoding layer. In Khamparia et al. (2020), a transfer learning process is developed for the diagnosis of detect breast cancer. A Modified VGG (MVGG) is developed for 2D and 3D images of mammograms. Coudray et al. (2018) designed a classification and mutation prediction model from non–small cell lung cancer histopathology images by the use of DL. In Asuntha and Srinivasan (2020), a DL based lung cancer detection and classification model is presented with distinct feature extraction techniques.

3 Materials and methods

In this study, the Glaucoma classification has been performed by the Multilayer Perceptron with CNN classification. The optic disc and optic cup boundary detected by Improvised CNN with entropy estimation. Further, the disc ratio, holistic and local features are extracted to avoid the interweavement with surrounding tissues and blood vessels.

The overall flow of the proposed system is shown in Fig. 2.

3.1 Segmenting the optic disc and cup boundary with entropy-based saliency detection block

Initially, the source retinal images are divided into numerous image blocks in which the entropy values are estimated. Then, the devised segmentation is structured using the CNN by using the alternative Conv layers, ReLU-Rectified Linear Unit layers, Decon Layer, and Softmax layer with the extraction of saliency features to improvise the human visualization. The weights are updated during the eye-retina optic disc and cup boundary segmentation operation. To obtain the final segmentation results, each pixel is classified and merged to get the results to form each sub-divided block. Saliency features are extracted primarily to establish good visualization for a variety of areas.

The convolutional layer is executing the dot product process intermediary to the filter and the source matrix (Li et al. 2021; Mansour, 2017, 2018; Mansour et al., 2021). This process aids in the reduction of the source image layer. ReLU layer is generated to vary the outcomes to optimization. The de-convolutional layer is generated to reverse the convolutional process for facilitating the additional smooth reduction of the dimensions and Softmax layer is introduced as the activating elements right before the classification layer. The steps for the CNN based segmenting operation are given below as follows:

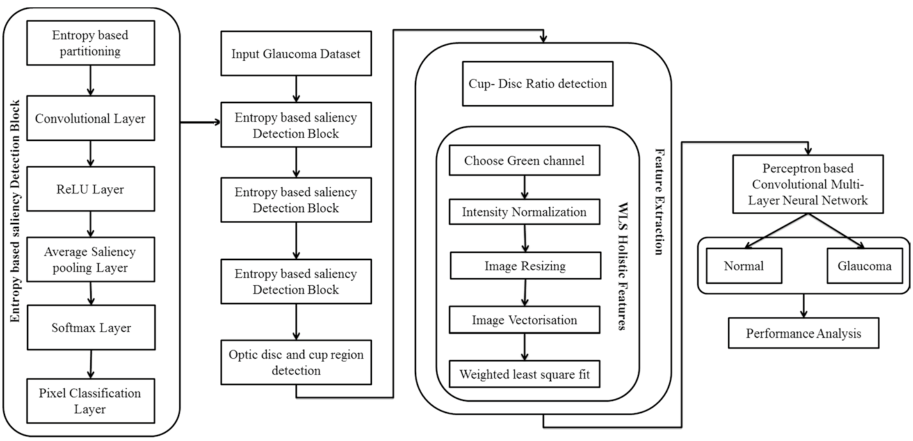


Fig. 2 Proposed flow for glaucoma detection

Steps for optic disc and cup boundary segmenting using CNN based entropy estimation

Algorithm 1

Input: Input Image

Output: Segmented Image

1. To ascertain the entropy for the image
$$E_{ntropy} = \sum_i X_i \log_2 X_j$$
2. $r \rightarrow E_{ntropy}\ row\ size$
3. $c \rightarrow E_{ntropy}\ column\ size$
4. $Initialize\ blocksize\ B_{size}$
   $for\ (\ i \rightarrow 1\ \ to\ (c - B_{size}) + 1)$
   $\quad for\ \ (j \rightarrow 1\ \ to\ (r - B_{size}) + 1)$
   $\quad\quad Par_{Imgae_a} \rightarrow\ E_{ntropy}(i{:}\, i + B_{size} - 1, j{:}\, j + B_{size} - 1)$
   $\quad\quad a = a + 1$
   $\quad end$
   end
5. Perform the convolution operation for each and every partitioned image
   $C_{conv}(r, c) \rightarrow \sum_{p \rightarrow -\infty}^{\infty} \sum_{q \rightarrow -\infty}^{\infty} H(p, q)\ \ Par_{Imgae}(r - p, c - q)$

Then, the updating of the weights is made by using the Maximum Likelihood Estimation (MLE) as shown below in the following steps.

Maximum likelihood estimation

Algorithm 2

1. Set $\delta_0 > 0, \varepsilon_0 > 0, \vartheta_\delta > 0, \vartheta_\varepsilon > 0$ & $Let\ \delta = \delta_0$
2. Maximum likelihood estimator $(\hat{\theta}, \hat{\rho})$ is computed by

$$(\hat{\theta}, \hat{\rho}) = \sum_{i=1}^{n} \log\left(\frac{1}{S_d}\sum_{j=1}^{J} h_p\right) \delta \sum_{i=1}^{n} \widehat{CV}\ (P_{wei})$$

Compute the estimated prediction error ε_γ

$\varepsilon_\delta = \frac{1}{nL}\sum_{l=1}^{L}\sum_{i=1}^{n}\left\| \hat{I}_{im}^{(l)}(t_i)\ \hat{I}_{im}(t_i)\right\|$

3. If $(\varepsilon_\delta < \varepsilon_0)$ then stop, otherwise, go to 4
4. Let $\delta_* = \delta\ \vartheta_\delta, compute\ \epsilon_{\delta_*}$
5. If $(\varepsilon_\delta \quad \epsilon_{\delta_*} > \vartheta_\varepsilon)$ then update $\delta = \delta_*$ and go back to step 3, otherwise, go to step 6
6. If $(\delta < \delta_0)$ then stop, otherwise, go to step 7
7. If $(\varepsilon_\delta < \varepsilon_0)$ then stop, otherwise, go to step 8
8. Let $(\delta_* = \delta + \vartheta_\delta)$ compute ϵ_{δ_*} as step 2
9. If $(\varepsilon_\delta\ \epsilon_{\delta_*} > \vartheta_\varepsilon)$ then update $\delta = \delta_*$ and go back to step 7, otherwise, stop

Where I_{im} Represents image, $P_{wei} \rightarrow weights$, S_d→ Sample standard deviation.

The maximum likelihood estimation MLE is defined as probability distribution parameters estimation using likelihood function maximization. The observed data is mostly possible based on supposed statistical model. The MLE is thus stated as parameter space point in which likelihood function is maximized. The MLE logic is flexible and intuitive. Maxima is determined by derivative test based on a differentiable likelihood function. The linear regression model likelihood maximized by ordinary least squares.

3.2 Weighted least square fit (WLS) holistic features

After the optic disc and cup boundary detection, the cup-disc ratio has been detected, the Green Channel G has been chosen due to the high contrast among the various structures of interest. Therefore, assume that the important analysis information of retinal fundus images has been provided. To the range between 0 and 255, all image intensities have normalized. To one-fourth of the number of its rows and columns, each gray-scale 2D image is then resized since the original set high dimensionality made recognition task's computer processing unfeasible. Finally, every image is in matrix form. Further, every image is transformed to n-dimensional row vector from its matrix form by organizing its elements with respect to column-wise order. Substantial computer resources are needed for the recognition of images due to the high dimensional image space. Before the recognition is performed, lower-dimensional spaces generally attempt recognition algorithms when maintaining the information as possible. In this study, weighted least square fit (WLS) is used.

Weighted least square is the linear regression and ordinary least squares generalization is referred as weighted linear regression in which it permits the co-variance matric errors are varied from matrix of identity. WLS is a generalized least squares specialization.

A generalized least square's special case namely weighted least squares happens when null = whole off-diagonal entries Ω, the observation variances are not equal still.

By the residual r_j, the data point to model fit is measured, explained as variation among the dependent variable of measured value y_j, the model predicted the value as, $f(x_i, \beta)$,

$$r_i(\beta) = y_i - f(x_i, \beta) \tag{1}$$

The minimum of function is estimated by equal variance of uncorrelated errors,

$$S(\beta) = \sum_i r_i(\beta)^2 \tag{2}$$

3.3 Perceptron based Convolutional Multi-Layer Neural network

At each pixel, the MLP and CNN predictive outputs are considered as n-dimensional vectors $L = (l_1, l_2, \dots, l_m)$, where the number of classes is n, every dimension $j \in [1, m]$ with respect to specific class probability as ($j^{th}class$) with specific membership relation. Preferably, the classification probability prediction is, for target class = 1 whereas for others = 0. Though due to the classification of remotely sensed image process's uncertainty, the probability value c is indicated as $f(y) = \{l_y \forall y \in (1{,}2, \dots, m)\}$, where $\sum_{y=1}^{m} l_y = 1$ and $l_y \in [0{,}1]$. The association of maximum membership basically takes the classification model as expected output label defined as class (L)

$$class(L) = argmax(\{f(y) = l_y \forall y \in (1{,}2, \dots, m)\}) \tag{3}$$

The membership association with confidence C is denoted as,

$$C = ma_l - me_l \tag{4}$$

$$ma_l = \max_{y \to 1\ tom} \left(l_y\right) \tag{5}$$

$$me_l = \frac{\sum_{y=1}^{m} l_y}{m} \tag{6}$$

From Eqs. (4–6), the maximum value of vector C represented as ma_l, whereas the average of all L values indicated as me_l. The difference among ma_l and me_l quantified C which calculates the class membership allocation confidence or reliability denoted as confidence map classification. The appropriate image patches have taken by CNN as its inputs as the substitute of image pixels and it exhibits the below things,

(1) At the central homogenous region, if the input image patch is situated, the purity of class is associatively high with higher variation among the predicted class membership relation compared with other classes and C value is likely to be higher.
(2) If the other land cover classes which are presents in the image patch as appropriate information, the prediction associated membership and with others distinction are said

to be low and C value tends to be small. Conversely, the perceptron of specific feature depends on spectral information per-pixel, thus preventing the membership association among classified objects of central and boundary regions.

Based on above things, the fusion decision rules are generated depends on CNN confidence. When its confidence is greater compared with predefined threshold (β_1), the output of fusion provides credit to CNN. The trusted perceptron provides the CNN confidence which is lesser compared with another threshold (β_2);

The fusion output selects the CNN or classification result of perceptron with greater confidence when the CNN confidence lies among the two thresholds ($\in(\beta_1, \beta_2)$). Thus, at location (r, c) forgiven image pixel, the class label $L(r, c)$ determined by rule-based decision fusion technique of the equivalent pixel measured as

$$L(r,c) = \begin{cases} Label_{PNet} C_{cnn} < \beta_1 \\ Label_{PNet} \beta_1 \leq C_{cnn} < \beta_2 and C_{cnn} < C_{PNet} \\ Label_{cnn} \beta_1 \leq C_{cnn} < \beta_2 and C_{cnn} > C_{PNet} \\ Label_{cnn} C_{cnn} \geq \beta_2 \end{cases} \tag{7}$$

From Eq. (7), the $Label_{PNet}$ and $Label_{cnn}$ denotes the Perceptron based Convolutional Multi-Layer Neural network classification results correspondingly. The classification confidence is C_{PNet} and C_{cnn} of Perceptron based Convolutional Multi-Layer Neural network.

4 Performance validation

The simulation of the presneted technqiue takes place using Python—3.6.5 in a PC Processor—i5-8600 k Graphics Card—GeForce 1050Ti 4 GB RAM—16 GB OS Storage—250 GB SSD File Storage—1 TB HDD. In addition, ten fold cross validation is applied to split the dataset into training and testing parts.

4.1 Dataset description

4.1.1 Image datasets

The study utilized two glaucoma diagnostic databases that are available publicly and comprise the retinal fundus image from RIM-One (https://www.idiap.ch/software/bob/docs/bob/bob.db.rimoner3/stable/index.html) and DRISHTI-GS (http://cvit.iiit.ac.in/projects/mip/drishti-gs/mip-dataset2/Home.php) database for the semantic pixel-wise segmentation of both OC and OD.

4.1.2 RIM-ONE

It is an open database of RF images and comprising 159 retina images. It comprised 74 glaucomatous and 85 healthy images of eyes. These images are extracted from three Spain hospitals and two ophthalmologists annotated the images.

4.1.3 DRISTI-GS

This dataset comprises 101 RF (retinal fundus) images captures with optical disc centered, dilated pupils, and 30 degree FOV. Collection and annotation of these images have been carried out by Aravind Eye clinic of Madurai. The size of these images is 2896 × 1944 pixel and preserved in uncompressed PNG file. This dataset provides average OC and OD borders on the basis of four professional modeling's.

4.2 Evaluation metrics

Performance analysis is carried out by calculating the values of various performance measures like accuracy, specificity, sensitivity, dice and overlap scores, and Jaccard coefficient for the OC and OD segmentation of proposed method associated with ground truth. These parameters are explained as,

$$Dice(DC) = \frac{2XTP}{(TP + FP) + (TP + FN)} \tag{8}$$

$$JaccardCoefficient(JC) = \frac{TP}{TP + FP + FN} \tag{9}$$

$$Sensitivity = \frac{TP}{TP + FN} \tag{10}$$

$$Specificity = \frac{TN}{TN + FP} \tag{11}$$

$$Accuracy = \frac{(TP + FN)}{(TP + FN + TN + FP)} \tag{12}$$

where TP, TN, FP, FN are the True Positive True Negative and False positive and False negative values correspondingly. Among segmented regions and ground truth, the overlapping error (E) and balanced accuracy (A) are measured. The E and A are expressed as,

$$E = 1 - \frac{Area(S \cup G)}{Area(S \cap G)} \tag{13}$$

From Eq. 13, the segmented region is S and G is the ground truth correspondingly.

$$A = \frac{1}{2}(Sensitivity + Specificity) \tag{14}$$

4.3 Results and discussion

The proposed method outperforms the existing results performed on Drishti-GS dataset shown in Table 1. Compared with the existing methods, the dice and overlap scores, Jaccard Coefficient, and Accuracy of proposed OC and OD segmentation show better results.

Table 1 Comparison of Existing methods (Al-Bander et al., 2018) and proposed method results for segmentation of OC and OD on Drishti-GS dataset

Ref	Techniques	Optic Cup OC			Optic Disc OD		
		DC (F)	JC(O)	Accu	DC(F)	JC(O)	Accu
Existing Methods (Al-Bander et al., 2018)	Regression model based Coupled shape	0.86	–	–	0.95	–	–
	CNN with Modified U-Net	0.85	0.75	–	–	–	–
	CNN based large pixel patch	0.93	0.87	–	–	–	–
	ISNT rule (Inferior Superior Nasal Temporal) and colour channel	–	0.97	–	–	–	0.99
	CNN with Ensemble learning	0.87	0.85	–	0.97	0.91	–
	Information based red channel	–	–	–	–	–	0.9454
	Fully convolutional based Dense Net model	0.82	0.71	0.99	0.94	0.90	0.99
Proposed	Proposed	0.99	0.97	0.98	0.99	0.98	0.99

Many of the existing systems used same Drishti-GS dataset for both training and testing. Further, in following Table 2, the proposed method outperforms the existing results performed on RIM-ONE dataset. Figures 2, 3 shows the OD and OC results of the proposed method with existing methods (Fig. 4).

The performance Drishti-GS shows better performance compared with RIM-ONE dataset for the segmentation of OC and OD. However, the proposed results are better than the existing results.

For Comp 2 (Table 3).

Above Table 3 and Fig. 5 shows the existing and proposed results associated with advanced solutions in case of detecting glaucoma. The cup-disc ratio CDR measurement is evaluated in terms of specificity and sensitivity and compared with the existing methods predicting the CDR values. The proposed method resulted from 100 sensitivity and 97.8 specificity.

For Comp 1 (Table 4).

From above Tables 4 and 5, it shows comparative analysis of proposed and existing method on Drishti and Rim-One datasets. In terms of specificity and sensitivity the proposed model is assessed. The proposed obtains sensitivity and specificity 95.32 and 100

Table 2 Comparison of Existing methods(Al-Bander et al. 2018) and proposed method results for segmentation of OC and OD on RIM-ONE dataset

Author	Techniques	Optic Disc OD		Optic Cup OC	
		DC (F)	JC(O)	DC(F)	JC(O)
Existing Methods(Al-Bander et al., 2018)	CNN with Modified U-Net	0.94	0.89	0.82	0.69
	Adversarial and Fully convolutional network	0.97	0.89	0.94	0.76
	Ant colony optimization	–	–	–	0.75
	Fully convolutional based Dense Net model	0.9	0.82	0.69	0.655
Proposed	Proposed	0.99	0.89	0.97	0.88

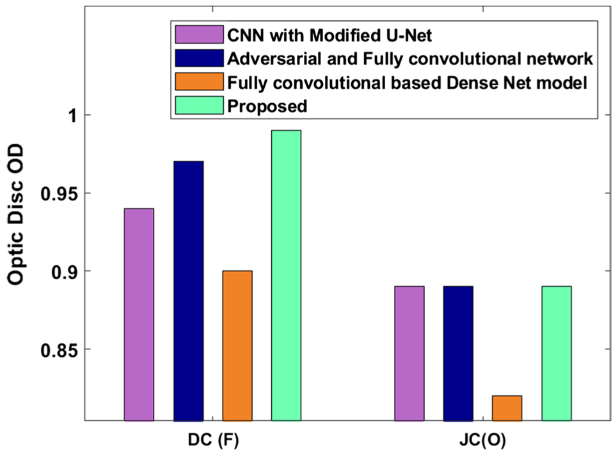


Fig. 3 Results analysis of proposed method on OD detection

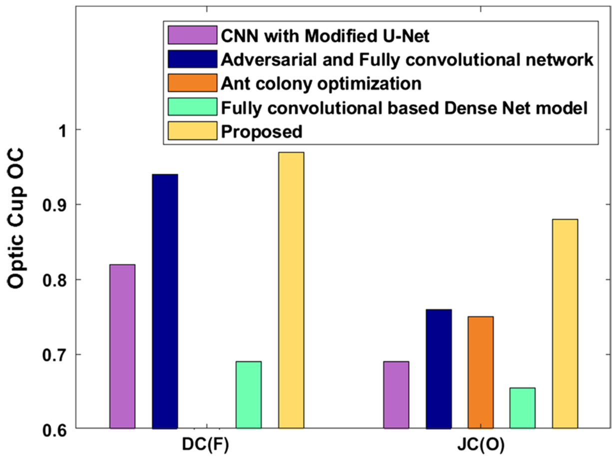


Fig. 4 Results analysis of proposed method with existing methods on OC detection

Table 3 Comparative analysis of the proposed with various existing methods (Fatima Bokhari et al. 2018) on DRISHTI dataset

Type of investigation	Specificity %	Sensitivity %
Stereo retinal images used for CDR	82	87
ISNT rule and CDR	80 to 99.2	97.6 to 100
CDR	82 to 94.7	60 to 94
Defects of retinal nerve fiber layer	54 to 75	80 to 90
Vessel bends and CDR	45 to 81.8	18.6 to 81.3
Glaucoma detection	97.5	98
Proposed	97.8	100

for OD and OC, the sensitivity and specificity 95.60 and 100 on Drishti and Rim-One.

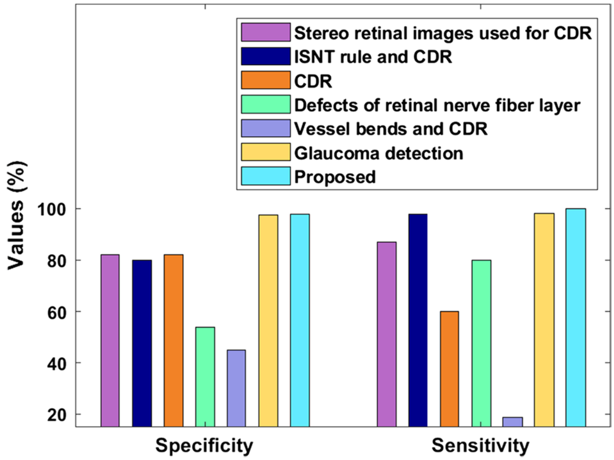


Fig. 5 Sensitivity and specificity analysis of proposed method with existing methods

Thus the useful insights from the results exhibited that the proposed method is highly desirable for OD and OC segmentation.

The AUC value obtained for the proposed method is 0.971 Drishti and it has compared with other existing methods and shown in Table 6. The higher AUC value indicates the highly accurate classification percentage of images for the whole system. Table 7 shows for OC and OD, AUC value separately as 0.96 and 0.974on Rim-One and 0.915 and 0.991 on Drishti correspondingly. The proposed model obtains better results compared with existing algorithms. For Drishti Dataset,

From the Tables 8, 9, clearly shows the input sample images, the optic disc OC and optic boundary OD which are detected precisely. By using the proposed perceptron based multi-layer classification the ratio of OC and OD boundary prediction is accurately performed.

For Rim-One dataset (Table 9).

5 Conclusion

A new segmentation method namely entropy estimation based CNN and Weighted Least Square Fit (WLS) Holistic Features is presented for accurately detect the optic disk and optic cup boundary OC and OD have been performed and the cup-disc ratio is measured. Further, the severity of Glaucoma has been detected using the Perceptron based Convolutional Multi-Layer Neural network for accurate prediction. For the comparative study, the proposed classification model is trained on the whole dataset with variations in parameters. The datasets used for analysis are the Drishti-GS and Rim-One dataset. The experimental results show the ratio of the OC and OD boundary predicted and depicted in the study. For future enhancement, diabetic retinopathy is classified based on this proposed algorithm in terms of its mild, moderate, and severe non-proliferative diabetic retinopathy NPDR.

Table 4 Comparison of proposed method OC and OD segmentation with existing method on Drishti-GS dataset (Tabassum et al., 2020)

Technique	OC				OD			
	DC (F)	JC (O)	Sensitivity	Specificity	DC (F)	JC (O)	Sensitivity	Specificity
Modified CNN	85.21	75.15	84.76	98.81	90.43	83.5	91.56	99.69
CNN with ensemble	87.1	85	–	–	97.3	91.4	–	–
Regression method	85	–	–	–	95	–	–	–
Locally statistical active counter method	84.7	–	–	–	95.5	–	–	–
Generative adversarial network	86.43	77.48	85.39	99.07	95.27	91.85	97.47	99.77
RNN based RACE ne	87	–	–	–	97	–	–	–
M–net	86.18	77.3	88.22	98.62	96.78	93.86	97.11	99.91
Patch based output space learning	85.8	–	–	–	96.5	–	–	–
Encoder-decoder net	88.18	80.06	88.19	99.9	96.42	93.23	97.59	99.9
U shaped CNN	89.12	82.30	–	–	97.8	94.9	–	–
CDED—net	92.4	86.32	95.67	99.81	95.97	91.83	97.54	99.73
Proposed	99.02	97.44	98.70	100	99.01	98.63	99.31	100

Table 5 Comparison of proposed method OC and OD segmentation with existing method on Rim-one dataset (Bharati et al., 2020)

Technique	OC				OD			
	DC (F)	JC (O)	Sensitivity	Specificity	DC (F)	JC (O)	Sensitivity	Specificity
Super pixel classification	74.4	73.2	–	–	82.93	82.93	–	–
Template based approach	–	–	–	–	84.2	84.2	–	–
Ant colony optimization	–	75.7	–	–	–	–	–	–
Modified U-net CNN	82	69	75.45	99.76	95	89	95.02	99.73
Ensemble CNN	82.4	80.2	–	–	94.2	89	–	–
Locally statistical active counter method	78.5	–	–	–	85.3	–	–	–
Generative adversarial network	82.5	71.65	81.42	99.65	95.32	91.22	94.57	99.87
M-Net	83.48	73	81.46	99.67	95.26	91.14	94.81	99.86
Encoder-decoder CE-Net	84.35	74.24	83.52	99.7	95.27	91.15	95.02	99.86
Patch based output space learning	78.7	–	–	–	86.5	–	–	–
U-shaped CNN	85.64	75.86	85.15	99.71	95.61	91.72	95.21	99.87
CDED -Net	86.67	75.32	95.17	99.81	95.82	91.01	97.34	99.73
Proposed	99.08	89.05	95.32	100	97.14	88.22	95.60	100

Table 6 AUC comparison on the Drishti dataset(Bharati et al., 2020)

AUC values	Method
0.7626	CNN (without vessel impainting)
0.7212	CNN (with vessel impainting)
0.82	Deep morphometric feature estimation
0.963	Encoder-Decoder based SegNet
0.971	Proposed system

Table 7 OC and OD segmentation results of the proposed system in accordance with AUC and Accuracy on Rim one and Drishti dataset

Method	Parameters	Rim one		Drishti	
		OD	OC	OD	OC
Existing	Accuracy	99.56	99.61	99.66	99.71
	AUC	0.987	0.909	0.969	0.957
Proposed	Accuracy	99.70	99.63	98.42	99.02
	AUC	0.991	0.915	0.974	0.96

Table 8 Optic Disc OD and Optic cup OC boundary detection on Drishti Dataset

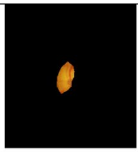
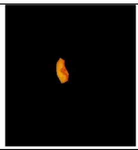
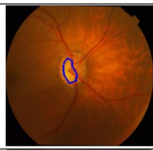
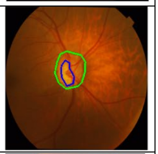
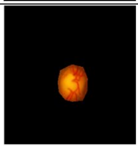
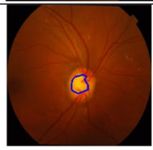
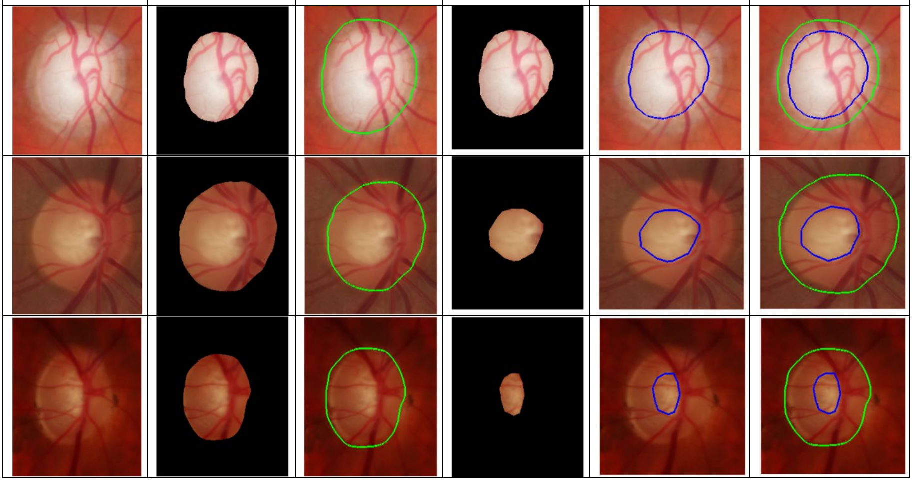
Input Image	Optic Disk Detection	Optic Disk Boundary Detection	Cup Area Detection	Cup Boundary Detection	Optic Disk and Cup Ratio Prediction
					
					
					

Table 9 Optic Disc OD and Optic cup OC boundary detection on Rim-One Dataset

Input Image	Optic Disk Detection	Optic Disk Boundary Detection	Cup Area Detection	Cup Boundary Detection	Optic Disk and Cup Ratio Prediction



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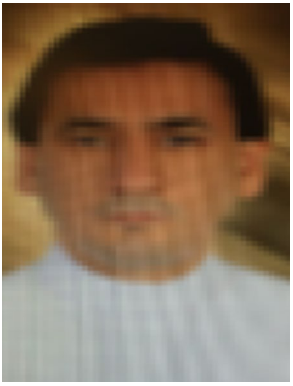

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